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Two-Phase Method for Image Authentication and Enhanced Decoding

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ABSTRACT A two-phase method for image authentication using side information from the communication channel, followed by improved error correction using reliable data, is presented. The functionality of the proposed method is divided into two phases. In the first phase, important fragments of an image, called the region of interest, such as an eye or face, are authenticated using side information from the channel. If the authentication succeeds, the second phase follows. In the second phase, the authenticated fragments are provided as a feedback to the channel decoder. This feedback is used to improve the results of channel decoding. By using a varying length of authenticated fragments at the receiver side, varying levels of coding gains are achieved. The first phase achieves limited image correction, together with the authentication of the region of interest. The second phase improves the decoding results using the image data, which are already declared correct and authentic by the first phase. The image authentication algorithm and error correcting codes mutually benefit from each other in the proposed method: the error correcting codes help in image authentication and successful image authentication helps in improving the results of error correcting codes. The applicability of the proposed scheme is supported by simulation results for images, where bit error rate curves are shown in addition to the visual image enhancements. A security analysis of the proposed method is given in the end.

INDEX TERMS Image authentication, coding gain, region of interest, turbo codes, known data, two phase scheme.

I. INTRODUCTION

A two phase method for image authentication and enhanced decoding is presented. In the first phase, soft authentication of image segments is performed. In the standard hard data authentication methods, such as those realized using MAC [1], [2], or HMAC [3], the data is declared unauthentic in the presence of even a single bit error. This is due to the well-known phenomenon called “Avalanche Effect” [4], where a difference of one or more bits at the input of a cryptographic algorithm results in a bit difference of an average of 50% at the output. Hard authentication is used to establish data origin authenticity and data integrity. Hard authentication can be improved using additional side information from the channel and/or channel decoder. Using this additional information, the decoding results can be improved and the data declared authentic. In some cases, such as in [5], in the presence of minor errors after the channel decoding, the authentication still succeeds if certain conditions are met, e.g., based on a threshold. In [6] the Hamming distance between the received authentication tag and the

recalculated tag is used as a threshold. This algorithm enables soft data authentication. The soft authentication method given in [5] authenticates data and performs error correction on the protected data using additional information from the channel and/or channel decoders such as SOVA [7] or MAP [8].

The second phase uses the authenticated fragments as “known data” to correct the remaining errors in the data. It is shown that this improves the decoding results and achieves a significant error recovery for the parts of the data out of the protected authenticated segments. As a result, not only robust image authentication is achieved, but the decoding results are also significantly improved as shown by the simulation results. In this paper, only image data is considered, but in general the results can also be other kinds of data such as audio and video data.

This paper is organized as follows. Section II presents related work. Section III presents the two phase method of authentication and improved decoding. Section IV presents the simulation results for the proposed two phase method.

A security analysis of the proposed method is given in Section V. Finally, the paper is concluded in Section VI.

II. RELATED WORK

Various hard and soft authentication algorithms have been proposed in literature. This includes, Soft Input Decryption (SID) [9], SID with threshold [6], Approximate Message Authentication Code (AMAC) [10], Approximate Image Message Authentication Code (AIMAC) [11], Noise Tolerant Message Authentication Code (NTMAC) [12] and its variants, Weighted NTMAC [13] and Error Correcting Weighted NTMAC [13]. Some of these authentication algorithms make use of channel measurements to improve the decoding results and then perform authentication, whereas the others are designed to be tolerant to certain modifications in the data.

SID uses the channel measurements or the reliability of data bits, the so called L-values or Log-likelihood Ratios (LLRs) [9] provided by a channel decoder to authenticate the data. For authentication, a standard hard data authentication method is used internally. Chase like decoding [14] is performed by the SID algorithm. The side information is used to correct data and the authentication succeeds only when the hard authentication succeeds. If the errors are not correctable, the data is declared unauthentic. This method works well for small data sizes and when the number of erroneous bits remaining after the channel decoder is low, e.g., less than 24 erroneous bits [5].

AMAC, on the contrary is a method of soft authentication, which achieves noise tolerant data authentication by tolerating certain errors using a threshold. If the original data is different from the data at the receiver, but the differences are below a threshold, the data is still accepted as authentic. If the number of errors is beyond the acceptable threshold, the data is declared unauthentic. AIMAC is a variant of AMAC in which AMAC is adapted such that it is tolerant to minor modifications in the image data. However, the intentional forgeries are still detectable. The capability of AIMAC against image modification due to JPEG compression, image forgery and additive Gaussian noise is discussed in [11]. The NTMAC algorithm [12], like AMAC, is tolerant to minor modifications in data. In NTMAC, the data is split into multiple blocks, and a standard MAC is calculated on each block. However, only a portion of the whole MAC is retained for each block, which is used for detecting modifications in that block. Partitions are introduced for tolerance to minor modifications. WNTMAC [13] is based on the NTMAC and introduces the concept of weights for differentiating the relatively important parts of data from the lesser important parts. NTMAC and WNTMAC do not provide error correction capability. EC-WNTMAC [13] is an extension of WNTMAC, where the error localization and correction capability is introduced aside from error tolerance.

It is noteworthy to point out that most of the above mentioned algorithms are based on the image data instead of the image content. There are numerous content based

authentication algorithms, which are also based on these algorithms for noise tolerant content authentication. An overview of the novel algorithms for robust image authentication is given in [15].

The idea of using “known data” to enhance the results of channel decoder is inspired from [16], where “dummy bits” were embedded in the transmitted data. The dummy bits can be all zeros or all ones and are known to the transmitter as well as the receiver. Since these bits are dummy, their values are not important by themselves, rather they are used to correct other bits of the data. The dummy bits are inserted in a zigzag manner and spread over the whole data. At the receiver side, their reliability values are set to $\pm\infty$. This can be done as the bits are known to have the value of zero or one, at both the transmitter and the receiver side, so the reliability of the bit being a zero or a one is 100%. The decoding is then performed using Maximum A Posteriori (MAP) decoder also known as BCJR algorithm after its inventors [8]. An improvement in the decoding results based on dummy bits is reported in [16]. However, the employed convolutional decoder is inferior in performance to Turbo codes [17], which uses multiple convolutional decoders in parallel.

The idea of combining image authentication and error correcting codes has been presented in literature. However, how the error correcting codes are actually used, varies from one approach to the other. For example, in the approach presented in [18], a robust two phase method for image authentication is presented, where error correcting codes are used for authentication tag generation and to introduce a certain degree of robustness in authentication. By protecting the image features using error correcting codes (such as Reed Solomon (RS) codes [19]), the features are at first tried for error correction and if there is success, the first phase of authentication succeeds. It is however important, that the algorithm works in two phases and the second phase must also succeed in order for the whole image to be declared authentic.

Other approaches include the use of error correcting codes in combination with watermarking, for error location / localization and correction. RS codes are used in [20] to generate parity symbols for every row and every column of the image data. The parity symbols are then embedded as a watermark in the two least significant bit planes of the image. The RS decoder at the receiver corrects the errors / modifications in the watermarked image and reconstructs the data of the original image. In [21], a watermarking approach is presented in which an image is divided into blocks and the block hash is encoded using a systematic error correcting code. The parity symbols are embedded into the blocks. During verification, the hash of each block is extracted from the embedded parity symbols, if the number of tampered blocks is below a predefined threshold.

The approach proposed in this paper is different from the above approaches as it authenticates an image in the first phase with the help of error correcting codes. Afterwards, the results of the authentication phase helps the error correcting codes in the second phase to improve the decoding results.

III. SOFT INPUT AUTHENTICATION AND IMPROVED TURBO DECODING

The proposed method uses the SID technique for image authentication, in combination with the Turbo decoder. The reliability information produced by the Turbo decoder is used in the authentication phase, i.e., the first phase. This phase contains “soft input authentication”, named in analogy to “soft input decryption”. When the first phase succeeds, the second phase uses the corrected data bits from the first phase as known bits, and feeds back this information to the Turbo decoder, so that the image correction can be improved further.

A. TRANSMITTER SIDE

The transmitter identifies a Region of Interest (RoI), such as an eye, face, etc., in the image, I , which should be authenticated. The RoI is extracted from the image I and a MAC tag, t , is calculated on the RoI using a secret key, k , shared between the transmitter and the receiver. The tag, t , is appended to the image I and the whole data is coded using Turbo codes. The encoded data is then stored or transmitted over a communication channel. The data might get altered on the storage medium or communications channel. It is noteworthy here, that the identification of the RoI is not covered by this work. It can be easily performed by appending the coordinates of the RoI to the image and transmitting these as a part of the encoded data. It can also be performed automatically at the transmitter and receiver by more advanced image processing techniques used for object detection and image segmentation, such as [22], [23]. In this paper, it is assumed that the location of RoI and the secret key k are pre-shared between the transmitter and the receiver. The transmitter side is shown in Fig. 1.

B. RECEIVER SIDE

The data received by the receiver might be different than the sent data depending on the errors due to channel noise during the data transmission and/or as a consequence of data storage. Without the loss of generality, only data transmission over a noisy channel is considered in this paper. The receiver decodes the received data using a Turbo decoder. The decoded data consists of I' and t' , which represent the potentially erroneous versions of the original image I and tag t .

This is due to the fact that depending on many factors, such as the level of noise at the channel, the decoder might not be able to correct all the errors. After decoding, the receiver operates in two phases as shown in Fig. 2, where the first phase authenticates the RoI and the second phase improves the decoding results using the data authenticated in the first phase.

1) PHASE-I

In phase-I, SID is performed on the RoI extracted from the decoded data. If SID on the RoI is successful, the second phase is executed. Otherwise, the data is declared unauthentic

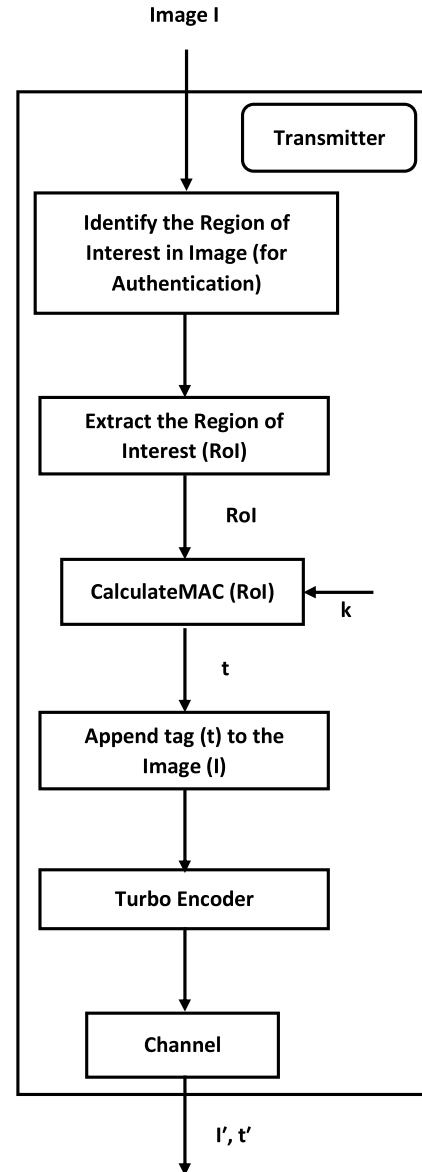


FIGURE 1. Transmitter side.

and the next phase is not executed. The phase-I of receiver starts after Turbo decoding as shown in Fig. 3.

The RoI is extracted from I' and denoted by RoI' , as it might be erroneous. A MAC tag t'' is calculated on the RoI' and compared with the received tag t' . If the tag verification fails, the SID iterations are executed. The SID considers the RoI' , the received tag t' , the recalculated tag t'' and the LLRs for the RoI' and t' [9].

Generally, the LLRs can be obtained from a decoder, such as the decoder for Turbo codes. However, in the absence of a decoder, the LLR values can also be obtained directly from the channel measurements, so called reliability values or L-values of the channel. Let the codeword $\mathbf{c} = (c_1, c_2, c_3, c_4, \dots, c_n)$, of length n , be transmitted over a noisy channel. The Soft Input Soft Output (SISO) channel

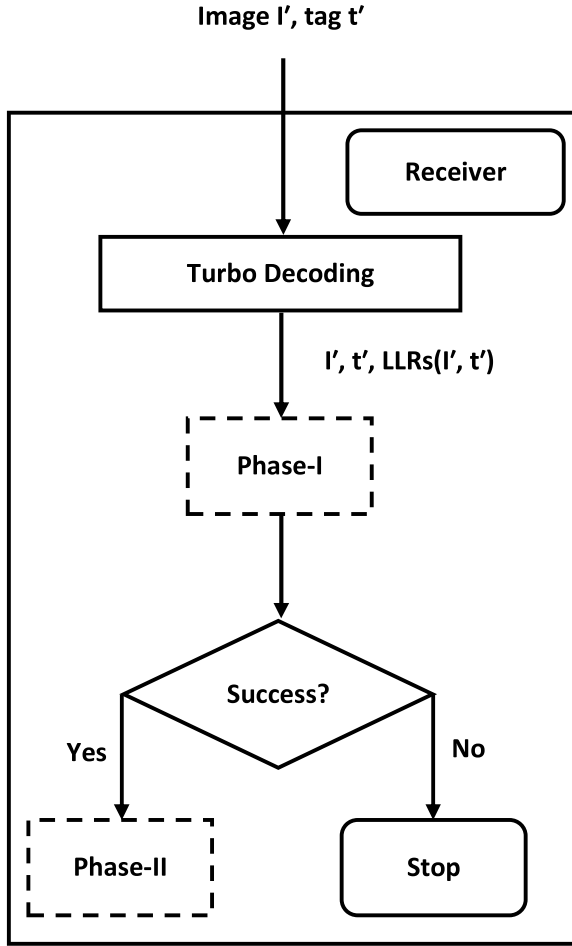


FIGURE 2. Receiver side.

decoder produces the LLRs as,

$$LLR(c_i) = \log \left(\frac{P(u_i = 1)}{P(u_i = 0)} \right), i = 1, 2, 3, \dots, n \quad (1)$$

LLRs are a measure of reliability of the decoded bits. The higher the absolute value of the LLR, the higher is the probability that the bit is correctly decoded. Vice versa, the lower the absolute LLR value, the lower is the probability that the bit value is correctly decoded. A SISO channel decoder, such as the Turbo decoder, produces an LLR value for each decoded bit.

The SID orders the bits of the RoI' according to their LLR values. The least reliable m bits are flipped iteratively in a certain combination, as in the Chase decoder [14]. In each iteration, a MAC tag, $\hat{t}$, is computed over the modified RoI''. The tag is compared with t' . If the two are equal, the authentication using SID is successful, otherwise the next combination of bits are flipped and the process is repeated. For further details please refer to [9]. The total number of iterations is dependent on the number of least reliable bits, m , used in the SID algorithm. Let N_r be the maximum number of iterations,

$$N_r = 2^m \quad (2)$$

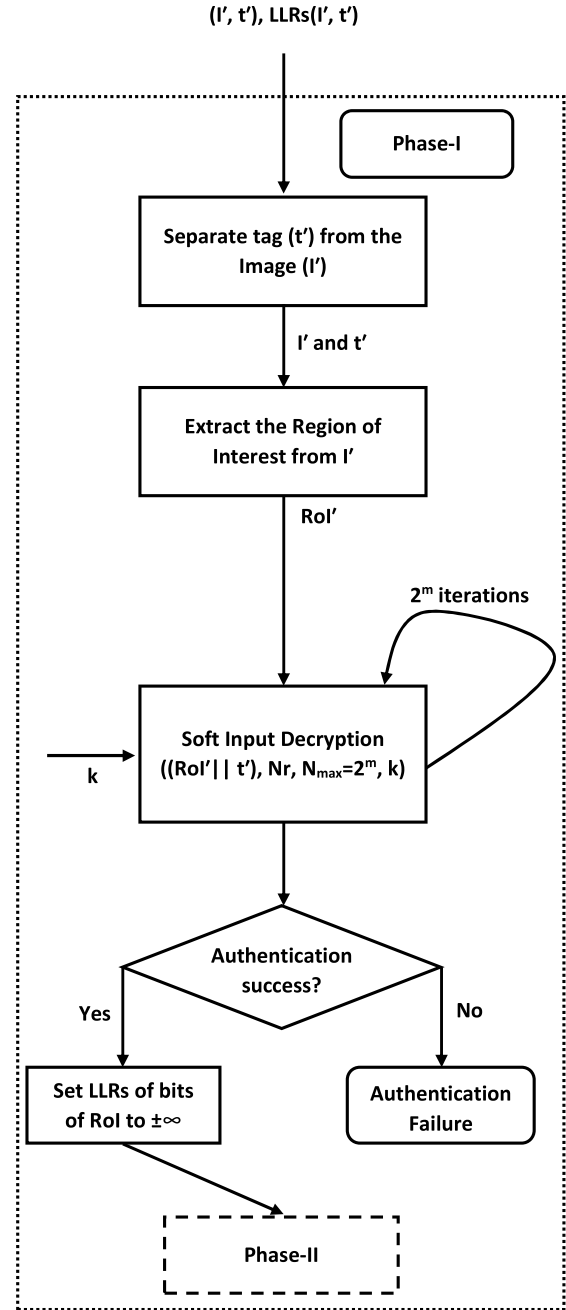


FIGURE 3. Soft Input Decryption during Phase-I.

When N_r iterations are performed and there is no successful authentication, the RoI' is declared as unauthentic. This means that the RoI' or its received tag t' , or both of them are erroneous. If the SID algorithm has found a valid RoI' and t' pair, i.e., when the RoI' and t' are corrected, the SID algorithm is successfully finished. Thus the RoI and t can be used as "known data" at the receiver for the second phase. The reliability values, LLRs, of these known data bits are set to $\pm\infty$ (i.e., to the theoretically maximum possible values) because it can be assumed that the data is authentic and sent by the transmitter with whom the receiver shares the secret key k . Thus, the data bits are fully reliable. These known data

bits are used to improve the decoding results of the Turbo decoder in phase-II. The pseudocode for soft input decryption is given in Algorithm 1.

Algorithm 1 Soft input decryption

Input:

- Received data and tag: $(x_{\text{received}} || t_{\text{received}})$
- Maximum number of iterations of SID: N_r
- Number of iterations of Turbo decoder: N_{max}
- Shared key: k

Output:

- Corrected data / tag

Procedure SID

$(x' || t')$, LLRs = Turbo_decoder($x_{\text{received}} || t_{\text{received}}$, N_{max})

If $(\text{MAC}(x', k) == t')$

$x'' = x'$

$t'' = t'$

Else

$sp = \text{Sort_bit_positions}(x' || t', \text{LLRs})$

$i = 1$

Do

{

$(x'' || t') = \text{Flip_next_bit_combination}$

$(x' || t', sp)$

If $(\text{MAC}(x'', k) == t'')$

break-while-loop

End if

$i = i + 1$

} While $(i \leq N_r)$

If $(i == N_r)$

$x'' = x'$

$t'' = t'$

End if

End if

return $(x'' || t'')$

end procedure

2) PHASE-II

Phase-II uses the authenticated RoI and its tag t as known data and through the corresponding LLRs, which are set to a maximum value, improves the decoding results of the Turbo decoder. The RoI, the tag t and their LLRs are provided as feedback to the Turbo decoder. At first, the RoI and t are spread pseudo-randomly over the whole image I' as shown in Fig. 4. The Turbo decoder is run again with the new input and LLRs. It is important to note that the Turbo decoder produces different results, depending on how many decoding iterations are chosen. This is shown in the section on simulation results. Thus different number of iterations can be chosen depending on the application. After Turbo decoding, the inverse permutation is used to spread the bits of RoI back to their original positions in I' .

The algorithm for enhanced error correction using known bits is given in Algorithm 2.

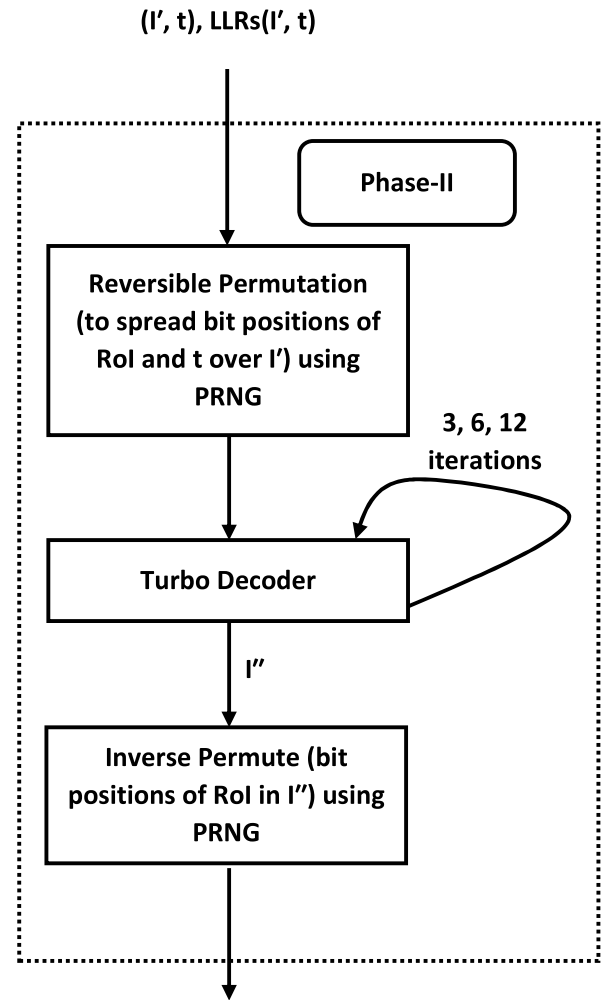


FIGURE 4. Enhanced decoding using known bits in Phase-II.

IV. SIMULATION RESULTS

A. GENERAL SIMULATION RESULTS

In this section, the simulation results are presented for various parameters of the proposed two phase method. Lena image of various resolutions, such as 32×32 , 64×64 and 128×128 are used in the simulations. Turbo codes of rate-1/3 are used and Additive White Gaussian Noise (AWGN) is considered as the channel model. The length of the MAC tag is chosen to be 160 bits. However, for a higher level of security, a larger length such as 256 or 512 bits should be chosen in practice. The maximum number of iterations for SID is chosen to be $N_r = 2^8$. The RoI is statically identified at both transmitter and receiver in these simulations, however, enhanced techniques for automatic recognition of the RoI can be used in practice, depending on the application.

The use of known bits helps in improving the Bit Error Rate (BER). Simulation results for different number of known bits are presented in Fig. 5-7. Fig. 5 presents the BER vs SNR results when three iterations of Turbo codes are performed. Different curves are shown for different number of known

Algorithm 2 Enhanced error correction using known bits**Input:**

- Received data: x'
- Known data: d
- Known data's positions: p
- Reliability of received data: LLRs
- Number of iterations of Turbo decoder: $N_{\max}$

Output:

- Data corrected using Turbo decoder with the help of known data

Procedure enhanced-error-correction-using-known-bits

Insert_known_data_bits_at_its_positions_in_received_data(x' , d , p)

max_value = MAX_VAL_OF_FLOATING_POINT;

Maximum value taken by a floating point number

Set_LLRS_at_known_positions_to_max(LLRs, p , max_value)

(x'' , LLRs') = Turbo_decoder(x' , $N_{\max}$);

Additional error detection code can be used here return x''

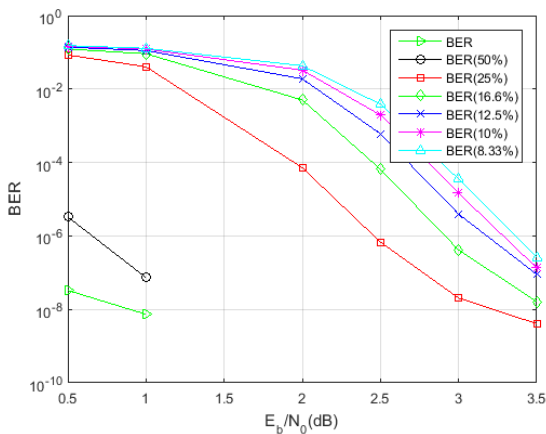
end procedure

FIGURE 5. BER vs SNR in the presence of known bits - three iterations of turbo decoder.

bits. The curve BER shows the bit error rate in the absence of known bits. The curve BER (50%) shows the case when 50% of the bits are known at the receiver, BER (25%) depicts the case when 25% of the bits are known at the receiver, and so on. Amongst the shown curves, the best BER curve is obtained for the case when 50% of the bits are known at the receiver and worst is for the case when 8.3% of the bits are known at the receiver.

When 50% of the bits are known, a coding gain of 2.5 dB is obtained at BER of 10^{-7} , as compared to the case when only 8.3% of the bits are known. These results are expected, as the higher number of known bits increases the probability of finding the correct decoding path in the Trellis diagram of Turbo decoder.

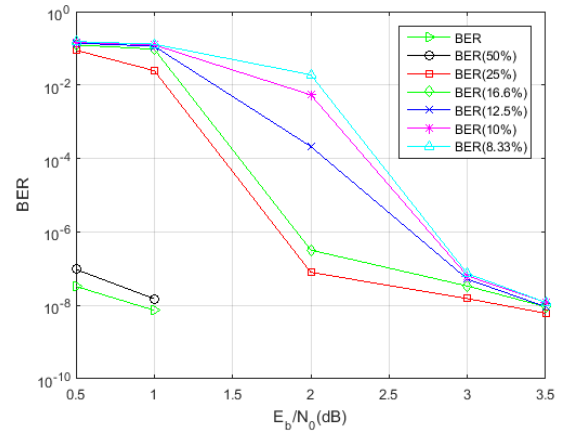


FIGURE 6. BER vs SNR in the presence of known bits - six iterations of turbo decoder.

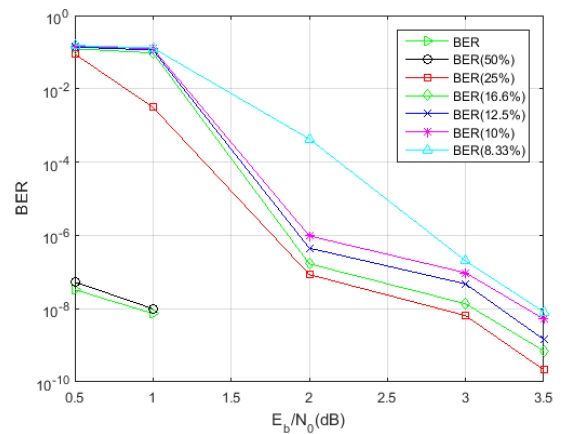


FIGURE 7. BER vs SNR in the presence of known bits - twelve iterations of turbo decoder.

The Fig. 6 shows the same results as in Fig. 5 but for six iterations of Turbo codes and Fig. 7 in case of twelve iterations. As expected, the decoding results improve with the increasing number of iterations of the Turbo decoder.

The results in Fig. 5-7 show the bit error rates as a measure of the quality of presented algorithm. However, for images, bit errors do not always result in much visual change, especially when the bit errors occur in the least significant bits (LSB) of the pixel values. Therefore, the corresponding results for the Lena image, which is used in simulations, are shown visually in Fig. 8-10. Various resolutions of Lena image are chosen, such as 32×32 , 64×64 and 128×128 pixels. For simplicity, Grayscale images are chosen for these simulations. Nevertheless, the results can be easily extended to colored images. The known bits are used, as discussed in Phase-II, to correct the other parts of the image.

Fig. 8 shows the Lena image in three different resolutions. For each resolution, phase-II of the proposed method is applied when 50%, 25%, 16% and 10% of the bits are known. The results are measured in the presence of AWGN channel at an SNR of 2.5 dB, and Turbo codes with three iterations. However, the improvement in the presence of

Resolution	Original image	50% bits are known at receiver	25% bits are known at receiver	16% bits are known at receiver	10% bits are known at receiver
32 × 32					
64 × 64					
128 × 128					

FIGURE 8. Enhanced error correction using known data with turbo codes of rate-1/3 and three iterations.

Resolution	Original image	50% bits are known at receiver	25% bits are known at receiver	16% bits are known at receiver	10% bits are known at receiver
32 × 32					
64 × 64					
128 × 128					

FIGURE 9. Enhanced error correction using known data with Turbo codes of rate-1/3 and six iterations.

known bits remain in the similar proportion in case of other SNR values.

Fig. 9 and Fig. 10 repeat the results shown in Fig. 8, but for six and twelve iterations of Turbo codes respectively. For a comparison, see Lena image of 128×128 pixels in Fig. 8, Fig. 9 and Fig. 10 respectively, when 10% of the bits are known at the receiver. The image recovery in the presence of the same level of noise is visually perfect under twelve iterations of Turbo decoder. In this particular run of the simulation, only 0.08% of the bits are left uncorrected after the twelfth iteration.

B. SIMULATION RESULTS OF THE ALGORITHM USING A SPECIFIC RoI

In this section, the application of the complete two-phase case is demonstrated on the original Lena image of resolution 128×128 pixels. In the example case, the right eye is chosen as the RoI. This RoI is extracted from the Lena image and

a MAC tag is calculated on it. The original image, and the MAC tag are passed through a Turbo encoder of rate 1/3. The encoded data is transferred over an AWGN channel at SNR of 2.5 dB.

At the receiver, the received data is decoded using a Turbo decoder of rate 1/3 and the RoI' and MAC tag t' are extracted. Soft Input decryption is performed as stated in phase-I of the proposed method. The verification succeeds (in case of the chosen noise value and the example parameters) and the eye (RoI) is declared authentic. For the image resolution of 128×128 pixels and a grayscale image where each pixel is represented by 8 bits, there are $128 \times 128 \times 8 = 131,072$ bits needed for the image representation. The RoI is 32×32 pixels and is represented by 8,192 bits which represent 6.25% of the whole image. Since the (potentially) known data at this stage is very low, the Turbo decoder is set to run 12 iterations. In cases where the SID algorithm succeeds and the RoI is declared authentic, the authenticated RoI bits are

Resolution	Original image	50% bits are known at receiver	25% bits are known at receiver	16% bits are known at receiver	10% bits are known at receiver
32 × 32					
64 × 64					
128 × 128					

FIGURE 10. Enhanced error correction using known data with turbo codes of rate-1/3 and twelve iterations.

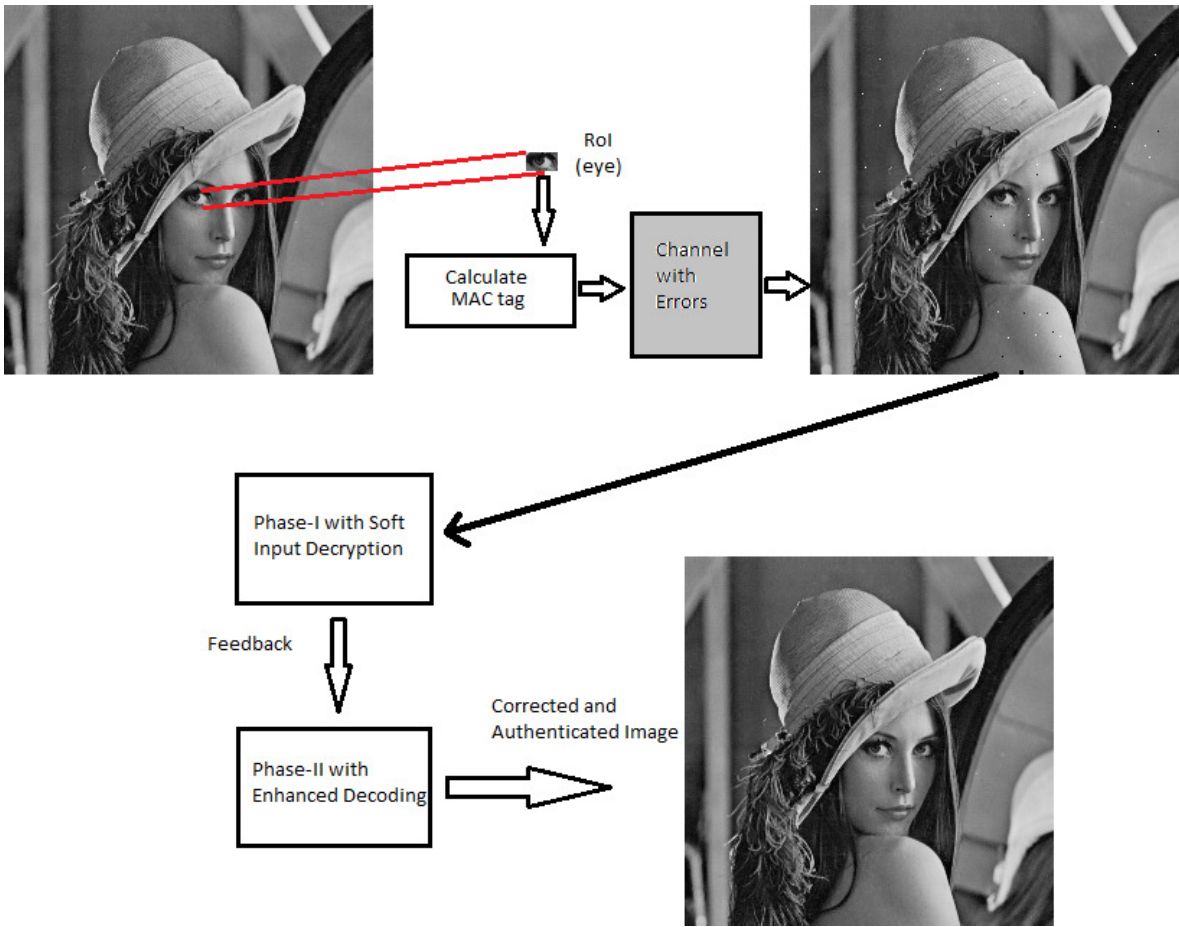


FIGURE 11. A case of successful image authentication and enhanced error correction.

treated as known data in the next phase for enhanced error correction.

The Phase-II starts with the “known data” generated by the phase-I. The LLRs of the known data bits are set to the

maximum value and spread over the whole image using invertible pseudorandom permutation. Turbo decoder is run again but now with the included known data bits, which are also randomly spread because of the pseudorandom

permutation. The results of the decoder show that the decoded data is the same as the original data, with no remaining errors. It should be noted that this might not be true for a lower SNR. The example is shown in Fig. 11 and it can be seen that no traces of errors are left in the resultant authentic image.

V. SECURITY ANALYSIS

False acceptance is the case when a wrong message is accepted as a correct message by the receiver. This occurs when the hash tag of a modified image (RoI in the case of this paper) matches the received hash tag. An attacker succeeds when a wrong image is accepted as correct by the system. An attacker can perform the forgery attack in the following two ways [5].

A. COLLISION BASED ON BIRTHDAY PARADOX

The attacker collects,

$$R = 1.17\sqrt{2^n}. \quad (3)$$

RoI / MAC pairs, where n is the length of the MAC tag and finds an external collision with a probability of 0.5 (based on the well-known Birthday paradox [24]).

B. COLLISION BASED ON RANDOM CHECKSUM

The attacker sends a message with a random checksum and hopes for a collision.

For an n bit MAC tag, the probability of a collision resulting in false acceptance of the message is,

$$P_{fa} = \frac{1}{2^n} \quad (4)$$

In this case, there are two further possibilities, i.e., there is a probability of false acceptance before the iterations of the SID algorithm, but after the SISO decoding (Case A) and then there is a probability of false acceptance during the iterations of the SID (Case B).

1) CASE A

The probability of false acceptance after the SISO decoding and before the iterative decoding of SID, depends on the BER after the SISO decoding, and is given by,

$$P_{fa} = (1 - (1 - BER)^m) \frac{1}{2^n} \leq \frac{1}{2^n} \quad (5)$$

2) CASE B

The probability of false acceptance after $(N_r - 1)$ iterations of the SID algorithm have been executed is given by,

$$P_{fa} = \frac{1}{2^n} \sum_{i=0}^{N_r-1} \left(1 - \frac{1}{2^n}\right)^i \quad (6)$$

Combining (5) and (6) gives the overall probability of false acceptance for the case when N_r verifications have been performed. This is given by,

$$P_{fa} = 1 - \left(1 - \frac{1}{2^n}\right)^{N_r} \quad (7)$$

If the length of the MAC is L bits and the number of iterations of SID algorithm is chosen to be $N_r = 2^m$, then the probability of false acceptance is increased from 2^{-L} (in case of standard hard verification) to 2^{-L+m} (in case of SID algorithm). As an example, in the simulations presented in Sec. IV, the length of MAC tag (L) is chosen as 160 bits and the number of iterations as $N_r = 2^8$. Therefore the P_{fa} is increased from 2^{-160} to 2^{-152} . This increase in P_{fa} can be compensated by increasing L (the length of MAC tag). However, this increase is the cost of the additional coding gain provided by the SID algorithm.

VI. CONCLUSION

A two phase method for image authentication and enhanced error decoding is presented. In the first phase, the region of interest in the received image is authenticated using soft input decryption technique. If the authentication succeeds, the second phase uses the authenticated data for improved decoding of the remaining data. Practical application of the proposed method is a scenario, where only part of an image needs to be authenticated, e.g., eye, face etc. A successful authentication is then used for additional improvement of the decoding results. Simulation results are presented for various parameters, showing a remarkable coding gain. A security analysis is provided, to show the impact of the proposed method on the probability of false acceptance. The paper shows how the synergy between the authentication mechanisms and decoding algorithms significantly improves the overall communication quality through an example of image application.

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